Eric Yang

eric.yang2@ucsf.edu | (310) 461-5746 | San Francisco, CA | Portfolio | Github | LinkedIn

Machine learning researcher and engineer specializing in AI evaluation, foundation models, and multimodal learning. M.S. in Health Data Science from UCSF with experience leading research across 156,724 medical images, LLM-to-physician benchmarking, and production AI systems spanning 100,000+ multimodal music tracks. Combines rigorous experimental design and failure analysis with end-to-end Python/SQL/PyTorch implementation.

EDUCATION

University of California, San Francisco (UCSF) — San Francisco, CA
M.S. in Health Data Science — Sept. 2026

University of California, Los Angeles (UCLA) — Los Angeles, CA
B.S. in Computational and Systems Biology, Concentration in Biological Data Sciences — Jun. 2024
Departmental Highest Honors

RESEARCH EXPERIENCE

UCSF Department of Medicine — San Francisco, CA
Data Science Specialist (Part-time) — Apr. 2025 – Apr. 2026

Empirical Evaluation of Large Language Models Against Human Expert Judgment

- Led empirical evaluation of GPT-5 Pro, GPT-4.1, and GPT-3.5 across ~500 medical-student OSCE encounters scored by 4 physician evaluators, benchmarking LLM-generated assessments against expert clinical judgment.
- Designed a cross-model evaluation framework using Cohen's κ, structured scoring, semantic similarity, and natural language inference; GPT-5 Pro achieved $\kappa = 0.67$, compared with 0.62 for GPT-4.1 and 0.60 for GPT-3.5 under the same evaluation protocol.
- Identified clinical organization and presentation structure as a major source of model-human disagreement: LLMs tended to over-credit generally coherent responses that physicians penalized for failing to follow clinically actionable presentation conventions.
- Built reproducible Python evaluation pipelines and translated failure analyses into rubric refinements for an AI-assisted medical simulation and feedback platform that provided students with rapid, structured performance feedback.

UCSF Hswen Lab — San Francisco, CA
Graduate Student Researcher — Oct. 2024 – Jun. 2026

Foundation Model Evaluation for Alzheimer's Disease Prediction

- Led first-author research evaluating retinal foundation models across 156,724 fundus images from 78,349 UK Biobank participants, including 535 AD cases.
- Designed a controlled benchmarking framework comparing RETFound, ViT-B16, ResNet-50, EfficientNet-B3, demographic baselines, multimodal fusion, and OCT-derived clinical models under consistent participant-level splits.
- Fine-tuned RETFound (ViT-Large) in PyTorch and developed end-to-end pipelines for image QC, preprocessing, class-imbalance mitigation, participant-level training, inference, and evaluation.
- Designed anatomical ablations across full-fundus, central-retinal, and vessel-enhanced representations; RETFound achieved AUROC 0.671 on full images versus 0.658 and 0.608, respectively.
- Demonstrated that fundus features added little incremental discrimination beyond demographics (AUROC 0.710 → 0.714), while OCT-derived measures increased AUROC to 0.795, consistent with stronger discriminative signal in OCT under the evaluated benchmark.
- Presented first-author findings at AAIC 2026; manuscript under review at Nature Medicine.

Systematic Analysis of Nicotine Pouch Discussions on Reddit

- Analyzed 37,797 Reddit posts using LDA topic modeling and VADER sentiment analysis, identifying distinct narratives around personal nicotine use, product endorsement, and availability; found predominantly positive sentiment (53.6% positive) alongside competing narratives of harm reduction, addiction, withdrawal, and adverse health effects, supporting a Social Influence manuscript under revision following peer review.

INDUSTRY EXPERIENCE

BOLERO Tokyo, Japan
Lead Data Scientist (Remote) Apr. 2025 – Mar. 2026

- Led a 6-person AI/data engineering team with end-to-end technical ownership of an internal music analytics platform, delivering the initial production demo in 3 weeks.
- Built multimodal ML pipelines across 100,000+ multilingual tracks, integrating librosa-derived audio features with metadata, behavioral, time-series, social-platform, streaming, and popularity signals.
- Developed and fine-tuned PyTorch multimodal neural networks for track similarity and genre-specific streaming, engagement, and longitudinal popularity modeling; used 10-fold cross-validation on the 70% training partition for model selection and evaluated final models on a held-out 15% test set.
- Built representation-learning and cosine-similarity retrieval pipelines against a reference database refreshed weekly with updated song-performance data, enabling new submissions to be benchmarked against dynamically evolving historical tracks.
- Developed a Llama 4 conversational assistant and web application used by 12+ employees and four music-group clients to upload tracks, retrieve comparable songs, inspect forecasts, and query results in natural language, reducing analytical reporting from approximately 2 days to minutes.
- Containerized and deployed the system with Docker and Microsoft Azure, integrating ML pipelines, databases, APIs, and backend services; owned full-stack implementation and production troubleshooting for a platform that remains in active use.
- Supported investor diligence and presentations with live platform demonstrations ahead of the company's ¥20M seed round.

PITTAN Kobe, Japan
Chief Technology Officer Jun. 2023 – Sep. 2023

- Led development of a non-invasive skin-analysis platform translating sweat amino-acid biomarker research into an end-to-end diagnostic and personalized skincare product.
- Built Python/Raspberry Pi 4 pipelines for remote liquid-chromatography control, data logging, peak processing, amino-acid quantification, and skin-condition scoring, collecting 500 measurements in 21 days.
- Developed neural-network models integrating sweat amino-acid and sensor-derived features from 500+ clinical samples for skin-condition assessment, achieving AUROC 0.85.

ADDITIONAL RESEARCH EXPERIENCE

UCLA & PITTAN Los Angeles, CA
Undergraduate Honors Thesis Jun. 2023 – Jun. 2024
Sweat Amino Acid Analysis for Non-Invasive Skin Assessment

- Independently designed and executed an experimental study evaluating sweat amino-acid biomarkers as a non-invasive diagnostic approach using 500+ clinical samples collected with Kobe University.
- Developed and evaluated random forest and multimodal neural-network models, assessing predictive performance, clinical feasibility, and translational potential.

UCLA Suthana Lab Santa Monica, CA
Research Assistant Mar. 2023 – Jun. 2024

- Processed 1,000+ EEG recordings and developed automated MATLAB signal-processing and machine-learning pipelines to improve analytical efficiency, reproducibility, and data quality.

MANUSCRIPTS

Yang, E., Buto, P., Vashisht, R., et al. (2026). *Limits of Fundus-Based Deep Learning for Alzheimer's Disease Classification: Anatomical Ablation and Clinical Benchmarking.* Under review at *Nature Medicine*.

Hswen, Y., **Yang, E.**, Han, E., et al. (2026). *Themes of Online Discussions of Nicotine Pouch Use.* Under revision following peer review at *Social Influence*.

PRESENTATIONS

Yang, E., Buto, P., Vashisht, R., et al. (2026). "Limits of Fundus-Based Deep Learning for Alzheimer's Disease Prediction: Anatomical Ablation and Clinical Benchmarking in UK Biobank." *AAIC 2026*, London, UK.

Yang, E., & Wong, F. (2026). “Simulating the Expert Evaluator: Immediate AI-Driven Feedback for Medical Student Oral Presentations and Written Documentation.” *UCSF AI & Education Symposium*, San Francisco, CA.

HONORS AND AWARDS

AAIC Conference Fellowship 2026
UCSF AGSD Summer Travel Award 2026

SKILLS

- **Programming:** Python, SQL, R, MATLAB, C/C++, Java, JavaScript
- **AI/ML:** PyTorch, scikit-learn, LLMs, Llama 4, Vision Transformers, RETFound, CNNs, Multimodal Neural Networks, Representation Learning, Fine-Tuning
- **Evaluation:** AUROC/AUPRC, Calibration, Bootstrapping, Ablation Studies, Model Benchmarking, Cohen's κ, Semantic Similarity, Natural Language Inference
- **ML Engineering:** Model Training & Inference, librosa, Audio Signal Processing, Cosine Similarity, Data Pipelines, APIs, Docker, AWS, TensorBoard, Git/GitHub